

JASON PHILLIP LU

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EDUCATION

University of Michigan <i>Doctor of Philosophy, Civil Engineering</i> Advisor: Prof. Neda Masoud Specialization in Next Generation Transportation Systems	<i>Expected 2028</i> GPA: 3.94/4.0
University of Michigan <i>Master of Science, Industrial and Operations Engineering</i>	<i>2026</i> GPA: 3.94/4.0
Georgia Institute of Technology <i>Bachelor of Science, Industrial Engineering</i> Research Advisor: Prof. Pascal Van Hentenryck Minor in Scientific and Engineering Computing	<i>2022</i> GPA: 3.93/4.0 <i>Highest Honors</i>

RESEARCH INTERESTS

Methodologies optimization, machine learning, game theory, simulation, network modeling
Applications shared mobility, connected and autonomous vehicles, logistics

PUBLICATIONS

Peer-Reviewed Journals

- Lu, J.**, Masoud, N. (2026). Incentive Design for Mobility Systems: A Review, Insights, and Future Directions. *Transportation Research Part E: Logistics and Transportation Review*. Volume 211, 104849. <https://doi.org/10.1016/j.tre.2026.104849>
- Lu, J.**, Trasatti, A., Guan, H., Dalmeijer, K., Van Hentenryck, P. (2024). The Impact of Congestion and Dedicated Lanes on On-Demand Multimodal Transit Systems [Special Issue: Multimodal Public Transport, Travel Behaviour and Social Equity]. *Travel Behaviour and Society*. Volume 36, 100772. <https://doi.org/10.1016/j.tbs.2024.100772>

Under Review

- Lu, J.**, Santanam, T., Guan, H., Riley, C., Kim, M.S., Trasatti, A., Masoud, N., Van Hentenryck, P. (2025). The Impact of Shared Autonomous Vehicles in Microtransit Systems: A Case Study in Atlanta. *Major Revision in Transportation Research Part C: Emerging Technologies*. <https://doi.org/10.48550/arXiv.2601.21046>
- Lu, J.**, Xia, X., Bahrami, S., Masoud, N. (2026). Path-Based Incentives for Multimodal Networks under Multiclass Traffic Assignment. *Major Revision in Transportation Research Part B: Methodological*. <https://doi.org/10.13140/RG.2.2.15344.62722>
- Lu, J.**, Masoud, N. (2026). The Incentive Bundle Program: Behaviorally-Consistent Incentive Design for Multimodal Transportation. *Submitted to Transportation Research Part B: Methodological*. <https://doi.org/10.13140/RG.2.2.22027.14882/1>

Working Papers

- Lu, J.**, Masoud, N. Cooperation and Competition in Multimodal Transportation Incentive Programs: A Tri-level Game-Theoretic Optimization Approach. *Extended Abstract Accepted at the 2026 INFORMS Transportation Science and Logistics Conference*.

Lu, J., Chen, C., Liu, Y., Wu, Y.L., Kusari, A., Masoud, N. A Large-Scale Optimization and Simulation Framework for Sensor Configuration in Connected and Autonomous Vehicles. *Abstract Accepted at the 2026 INFORMS Annual Meeting.*

PRESENTATIONS

Invited Talks

A Large-Scale Optimization and Simulation Framework for Sensor Configuration in Connected and Autonomous Vehicles. *Next Generation Transportation Systems Seminar Series*, Department of Civil and Environmental Engineering, University of Michigan, March 19, 2026.

The Incentive Bundle Program to Enhance Multimodal Transport. *Next Generation Transportation Systems Seminar Series*, Department of Civil and Environmental Engineering, University of Michigan, February 27, 2025.

Conference Presentations

Lu, J., Masoud, N. Cooperation and Competition in Multimodal Transportation Incentive Programs: A Tri-level Game-Theoretic Optimization Approach. *2026 INFORMS Transportation Science and Logistics Conference*, Boston, Massachusetts, July 26-29, 2026.

Masoud, S., Masoud, N., Alinezhad, E., **Lu, J. (presenter)**, Rapp, S., Rimanelli, J. Enhancing Military Digital Twins: Leveraging Dynamic Data-Driven Application Systems for Complex Operational Scenarios. *2025 ARC Annual Program Review*, Ann Arbor, Michigan, June 17-18, 2025.

Lu, J., Chen, C., Liu, Y., Wu, Y.L., Kusari, A., Masoud, N. A Large-Scale Optimization and Simulation Framework for Sensor Configuration in Connected and Autonomous Vehicles. *2026 CCAT Global Symposium on Mobility Innovation*, Ann Arbor, Michigan, April 17, 2026.

Masoud, N., **Lu, J.**, Chen, C., Liu, Y., Wu, Y., Kusari, A. Balancing Safety, Cybersecurity, and Mobility: Quantifying the Impact of Sensor Redundancy in Connected and Autonomous Vehicles. *2026 CCAT Global Symposium on Mobility Innovation*, Ann Arbor, Michigan, April 17, 2026.

Lu, J., Xia, X., Bahrami, S., Masoud, N. Can Incentives Reshape How We Move? A Multimodal Traffic Assignment Perspective. *The 105th Transportation Research Board Annual Meeting*, Washington, D.C., January 11-15, 2026.

Lu, J., Masoud, N. A Game Between Ridehailing Operator and Government Under the Incentive Bundle Program. *2025 INFORMS Annual Meeting*, Atlanta, Georgia, October 26-29, 2025.

Lu, J., Masoud, N. The Incentive Bundle Program to Enhance Multimodal Transportation. *The 7th Bridging Transportation Researchers Conference*, Virtual, August 6-7, 2025.

Masoud, S., Masoud, N., Alinezhad, E., **Lu, J. (presenter)**, Rapp, S., Rimanelli, J. Adaptive Contested Logistics: A Dynamic Data-Driven Application System Framework with Dynamic Bayesian Belief Networks. *2025 ARC Annual Program Review*, Ann Arbor, Michigan, June 17-18, 2025.

Masoud, N., **Lu, J.**, Chen, C., Liu, Y., Wu, Y., Kusari, A. Balancing Safety, Cybersecurity, and Mobility: Quantifying the Impact of Sensor Redundancy in Connected and Autonomous Vehicles. *2025 CCAT Global Symposium on Mobility Innovation*, Ann Arbor, Michigan, March 28, 2025.

Lu, J., Masoud, N. Designing Incentive Bundles for Multimodal Transportation Systems. *2024 INFORMS Annual Meeting*, Seattle, Washington, October 20-23, 2024.

Akhlaghi, V.E., Guan, H., **Lu, J.**, Van Hentenryck, P. Optimizing Truck Fleet Scheduling for Fuel Deliveries. *2023 INFORMS Annual Meeting*, Phoenix, Arizona, October 15-18, 2023.

ADVISING

Masters Students

Xingpeng Xia, University of Michigan
M.S. Civil Engineering (Expected 2026)

Yuyang Liu, University of Michigan
M.S. Electrical Engineering (2026)
Placement: PhD Student, University of Michigan

Undergraduate Students

Tianlang Chen, University of Michigan
B.S. Data Science (Expected 2027)

Yi Ling Wu, University of Michigan
B.S. Computer Engineering (2025)
Placement: PhD Student, Texas A&M University
– Aviles-Johnson Doctoral Fellowship

TEACHING

Graduate Student Instructor

University of Michigan

CEE 553 - Infrastructure Systems Optimization
Instructor: Prof. Neda Masoud

Fall 2026

Guest Lecturer

University of Michigan

Gurobi
CEE 553 - Infrastructure Systems Optimization

Fall 2024, Fall 2025

Overview of Next Generation Transportation Systems Program
CEE 200 - Introduction to Civil and Environmental Engineering

Spring 2025

Preparing for the Civil and Environmental Engineering Career Fair
CEE 200 - Introduction to Civil and Environmental Engineering

Fall 2024

Head Teaching Assistant

Georgia Institute of Technology

ISYE 4134 - Constraint Programming
Instructors: Mr. Tejas Santanam, Prof. Pascal Van Hentenryck

Spring 2023

Undergraduate Teaching Assistant

Georgia Institute of Technology

ISYE 3044 - Simulation Analysis and Design
Instructor: Prof. Seong-Hee Kim

Summer 2022

ISYE 4034 - Decision and Data Analytics
Instructor: Prof. Jye-Chyi Lu

Spring 2022

ISYE 3770 - Statistics and Applications
Instructor: Dr. Tuba Ketenci

Spring 2021

HONORS AND AWARDS

Western Track Best Paper Award
Bridging Transportation Researchers Conference

2025

Rackham Conference Travel Grant
Rackham Graduate School, University of Michigan

2024, 2025

Seth Bonder Fellowship*2022, 2023*

H. Milton Stewart School of Industrial and Systems Engineering, Georgia Institute of Technology

Stewart Topper Fellowship (Declined)*2023*

H. Milton Stewart School of Industrial and Systems Engineering, Georgia Institute of Technology

Senior Design Capstone Finalist*2022*

H. Milton Stewart School of Industrial and Systems Engineering, Georgia Institute of Technology

SERVICE - INTERNAL

Michigan Transportation Student Organization

Vice President

*May 2024 - Apr. 2025***The Seth Bonder Camp in Computational and Data Science for Engineering**

Instructor

*Jun. 2022 - Jul. 2023***SERVICE - EXTERNAL**

Reviewer for Academic Journals and Conferences

- Transportation Research Board
- Bridging Transportation Researchers

Conference Session Chair

- Advancements in Urban Transit Systems, *2024 INFORMS Annual Meeting*, Seattle, Washington, October 20-23, 2024.

INDUSTRY EXPERIENCE

Convoy

Process Improvement Intern

*Jan. 2022 - May 2022***Yokogawa**

Industrial Engineering Intern

May 2021 - Dec. 2021

Industrial Engineering Intern

May 2020 - Jul. 2020